15. Iris Flower Classification using Naïve Bayes Classifier

Aim

To implement **Iris Flower Classification** using the **Naïve Bayes classifier** in Python and predict the species of iris flowers.

Algorithm

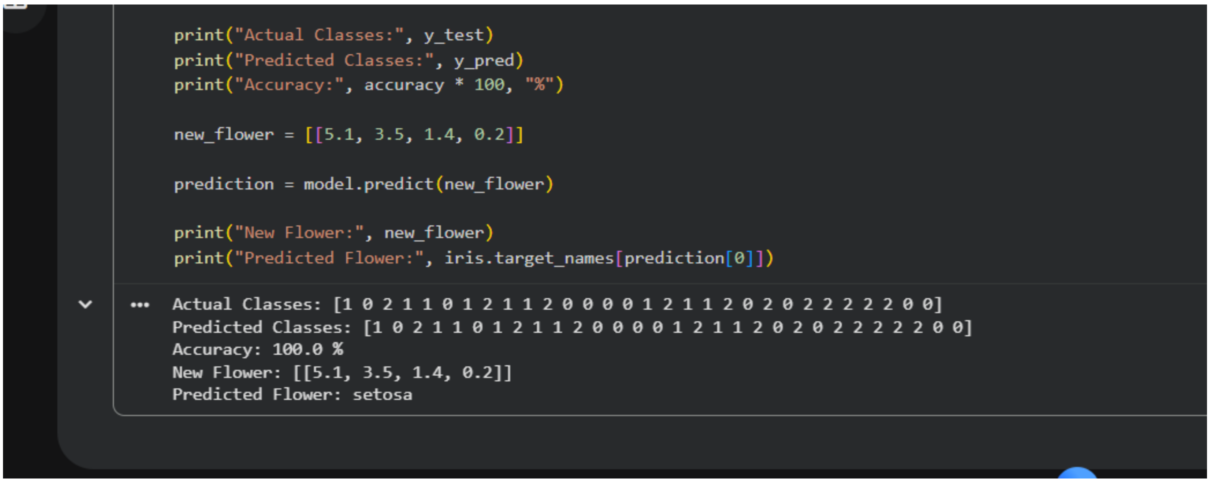
1. Import the required libraries.
2. Load the Iris dataset.
3. Separate the input features and target classes.
4. Split the dataset into training and testing data.
5. Create a Gaussian Naïve Bayes classifier.
6. Train the classifier using the training data.
7. Predict the classes of the test data.
8. Calculate the accuracy of the model.
9. Predict the class of a new iris flower.

Code:

```
from sklearn.datasets import load_iris

from sklearn.model_selection import train_test_split

from sklearn.naive_bayes import GaussianNB

from sklearn.metrics import accuracy_score


iris = load_iris()


X = iris.data

y = iris.target


X_train, X_test, y_train, y_test = train_test_split(

    X, y, test_size=0.2, random_state=42

)


model = GaussianNB()
```

```
model.fit(X_train, y_train)

y_pred = model.predict(X_test)

accuracy = accuracy_score(y_test, y_pred)

print("Actual Classes:", y_test)
print("Predicted Classes:", y_pred)
print("Accuracy:", accuracy * 100, "%")

new_flower = [[5.1, 3.5, 1.4, 0.2]]

prediction = model.predict(new_flower)

print("New Flower:", new_flower)
print("Predicted Flower:", iris.target_names[prediction[0]])
```

OUTPUT:



Result Statement

Thus, **Iris Flower Classification using the Naïve Bayes classifier** was successfully implemented in Python, and the species of a new iris flower was successfully predicted.